

Clinical ECG: Conduction blocks, Pacemakers, and Electrolyte Disturbances

Ermin Tale DO, Internal Medicine PGY-1
Jacobi Medical Center/Albert Einstein College of Medicine
ermin.tale01@gmail.com

Todd Cohen MD, Chief of Cardiology and Director of Medical
Device Innovation
tochen03@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

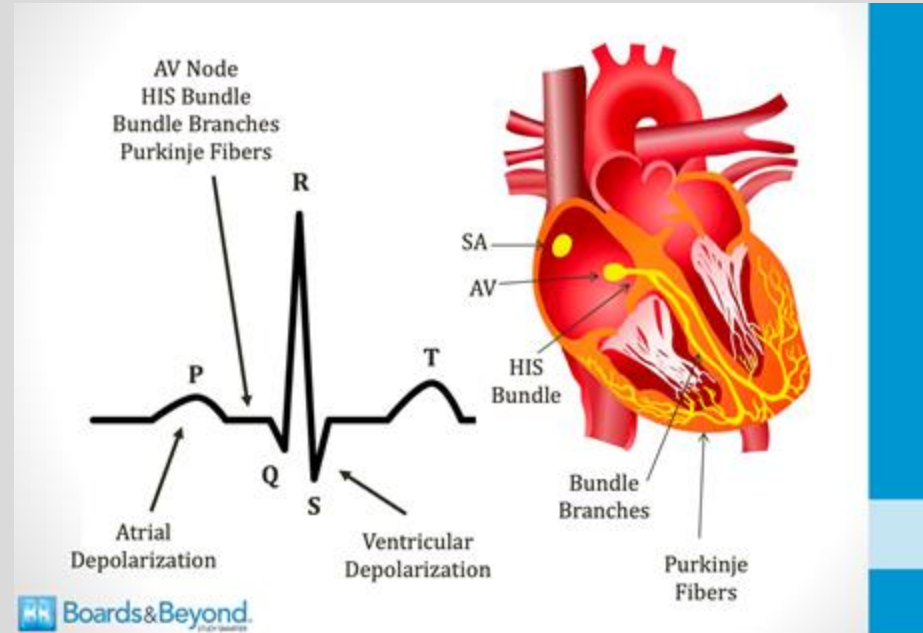
Session Objectives

1. Differentiate and manage the different types of AV blocks.
2. Understand bundle anatomy and identify between left vs right bundle branch block.
3. Recognize ECG changes due to electrolyte imbalances.

Heart or AV Conduction Blocks (same thing)

- Delay or interruption in the transmission of the impulse from the atria to the ventricles
- Two locations: AV node and His-Purkinje system

Disorder	Common Site of Block
1 st Degree	AV node
Mobitz I	AV Node
Mobitz II	His-Purkinje System
3 rd degree (Complete)	His-Purkinje System



The Heart Block Poem

- If the R is far from the P, then you have a 1st Degree



- PR gets longer, longer, longer, drop, it's a case of Wenckebach!



- If some R's don't get through, prepare to pace that Mobitz II!



- If the R's & P's don't agree, prepare to PACE that 3rd degree!



Least Severe



Most Severe

1st
Degree

2nd
Degree
Type 1

2nd
Degree
Type 2

3rd
Degree

First Degree AV Heart Block

- Delayed conduction from atrium to ventricle (but no interruption)
- **Prolonged PR interval** (> 200 msec)
- All PR intervals are equal and all P waves Conducted (QRS follows)
- Usually **asymptomatic**
- Can be caused by medications
- **No treatment required**



The Heart Block Poem

- If the R is far from the P, then you have a 1st Degree



- PR gets longer, longer, longer, drop, it's a case of Wenckebach!



- If some R's don't get through, prepare to pace that Mobitz II!



- If the R's & P's don't agree, prepare to PACE that 3rd degree!



Least Severe



Most Severe

1st
Degree

2nd
Degree
Type 1

2nd
Degree
Type 2

3rd
Degree

Second Degree AV Block: Mobitz Type I (Wenckebach)

- The **PR interval progressively increases** until one p-wave is blocked and isn't transmitted to the ventricles
- Notable **PR interval lengthening with a P-wave not followed by a QRS complex**
- The cycle starts all over again after that drop (regularly irregular)
- **Treatment is usually not indicated** as this rhythm usually is asymptomatic



The Heart Block Poem

- If the R is far from the P, then you have a 1st Degree



- PR gets longer, longer, longer, drop, it's a case of Wenckebach!



- If some R's don't get through, prepare to pace that Mobitz II!



- If the R's & P's don't agree, prepare to PACE that 3rd degree!



Least Severe



Most Severe

1st
Degree

2nd
Degree
Type 1

2nd
Degree
Type 2

3rd
Degree

Second Degree AV Block: Mobitz Type II

- Dropped QRS complex that is **NOT** preceded by a lengthening of the PR interval
- The **PR interval is consistent** (either normal or prolonged), until one p-wave is blocked and isn't transmitted to the ventricles
- Symptoms may be present and are more common than Mobitz Type I and include presyncope, syncope, fatigue, or SOB
- More serious than type I second-degree AV block
- May progress to third degree (complete) AV block
- Often **treated with a pacemaker**



The Heart Block Poem

- If the R is far from the P, then you have a 1st Degree



- PR gets longer, longer, longer, drop, it's a case of Wenckebach!



- If some R's don't get through, prepare to pace that Mobitz II!



- If the R's & P's don't agree, prepare to PACE that 3rd degree!



Least Severe



Most Severe

1st
Degree

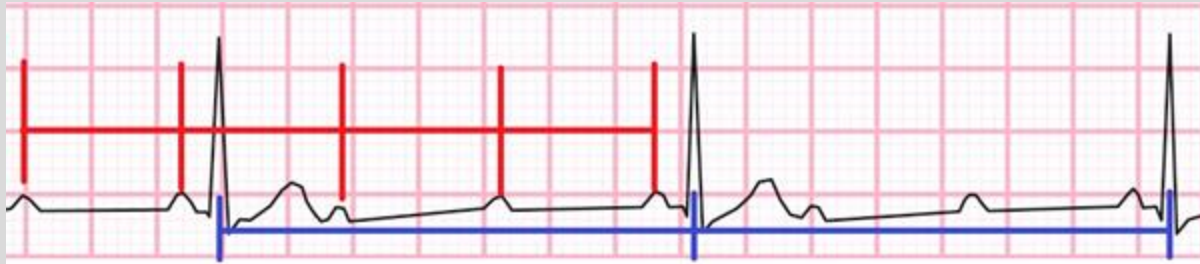
2nd
Degree
Type 1

2nd
Degree
Type 2

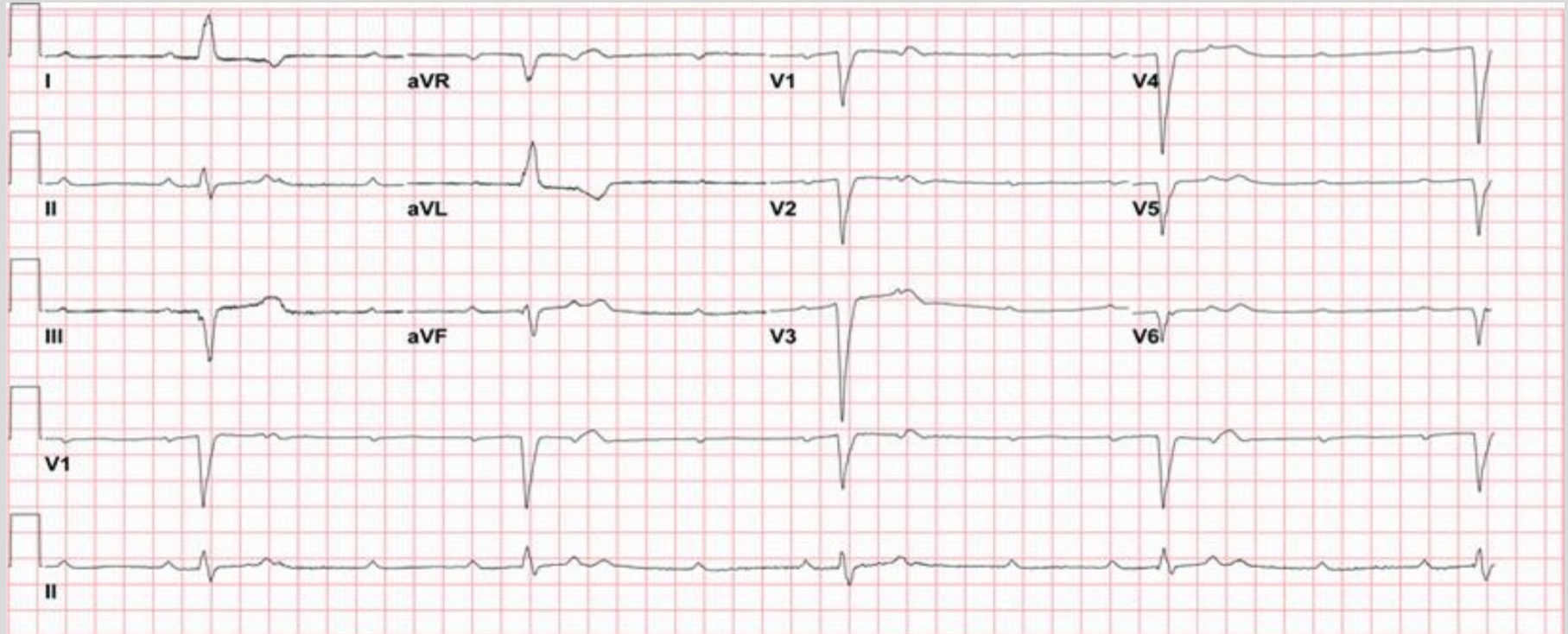
3rd
Degree

Third Degree (Complete) Heart Block

- The **atria and ventricles beat independently** of each other
- P-waves are **not associated** with QRS complexes
- **Atrial pacing is greater than ventricular pacing**
- ECG will show **equal RR intervals** and **equal PP intervals** but no association
- May be caused by MI, degeneration of the conductive tissue, and **Lyme Disease**
- Treated with a **Pacemaker**



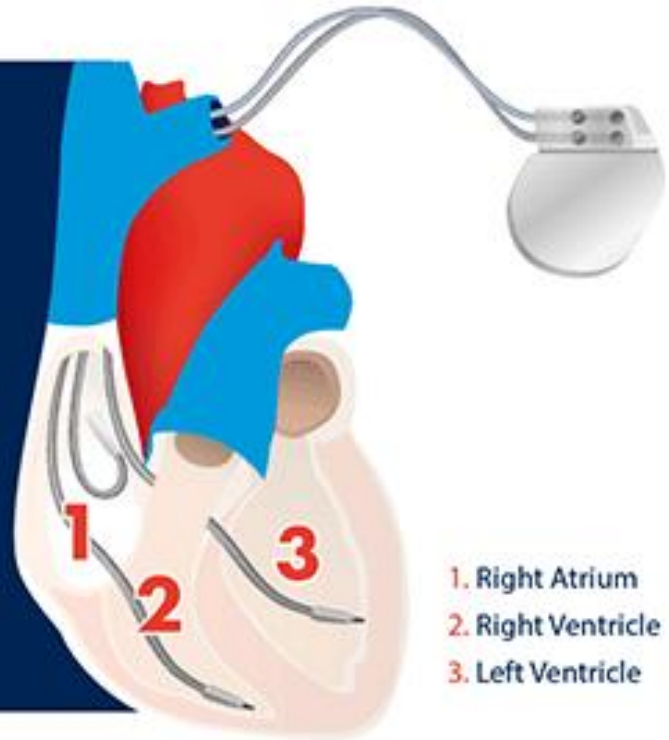
Third Degree (Complete) Heart Block



Pacemaker

Single Chamber Pacemaker

Uses one lead in the upper chambers (1) OR
one lead in the lower chambers (2) of the heart.



- 1. Right Atrium
- 2. Right Ventricle
- 3. Left Ventricle

Summary: AV Blocks

Site of Block:*

AV node

AV node

His-Purkinje

His-Purkinje

First degree AV block



Second degree AV block (Mobitz I or Wenckebach)



Second degree AV block (Mobitz II)



Third degree AV block



The Heart Block Poem

- If the R is far from the P, then you have a 1st Degree



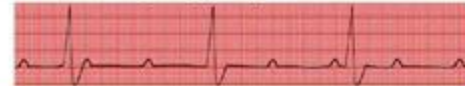
- PR gets longer, longer, longer, drop, it's a case of Wenckebach!



- If some R's don't get through, prepare to pace that Mobitz III!



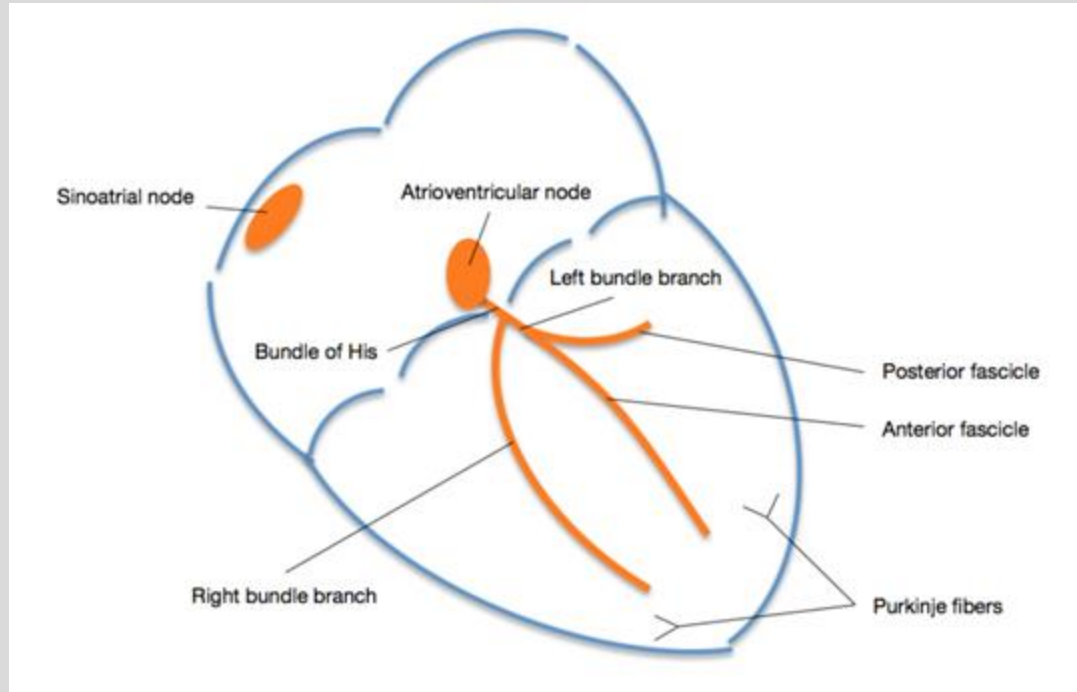
- If the R's & P's don't agree, prepare to PACE that 3rd degree!



Heart Block Poem

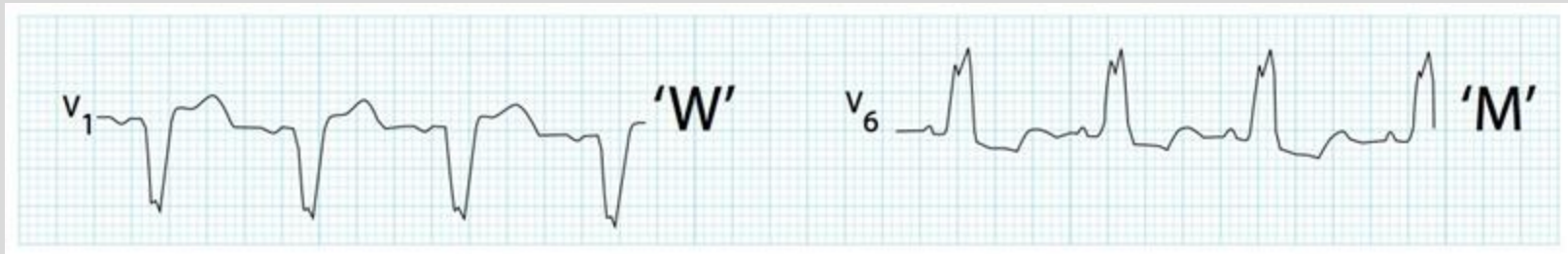
*Least to most severe

Cardiac Conduction System: Bundle Branches

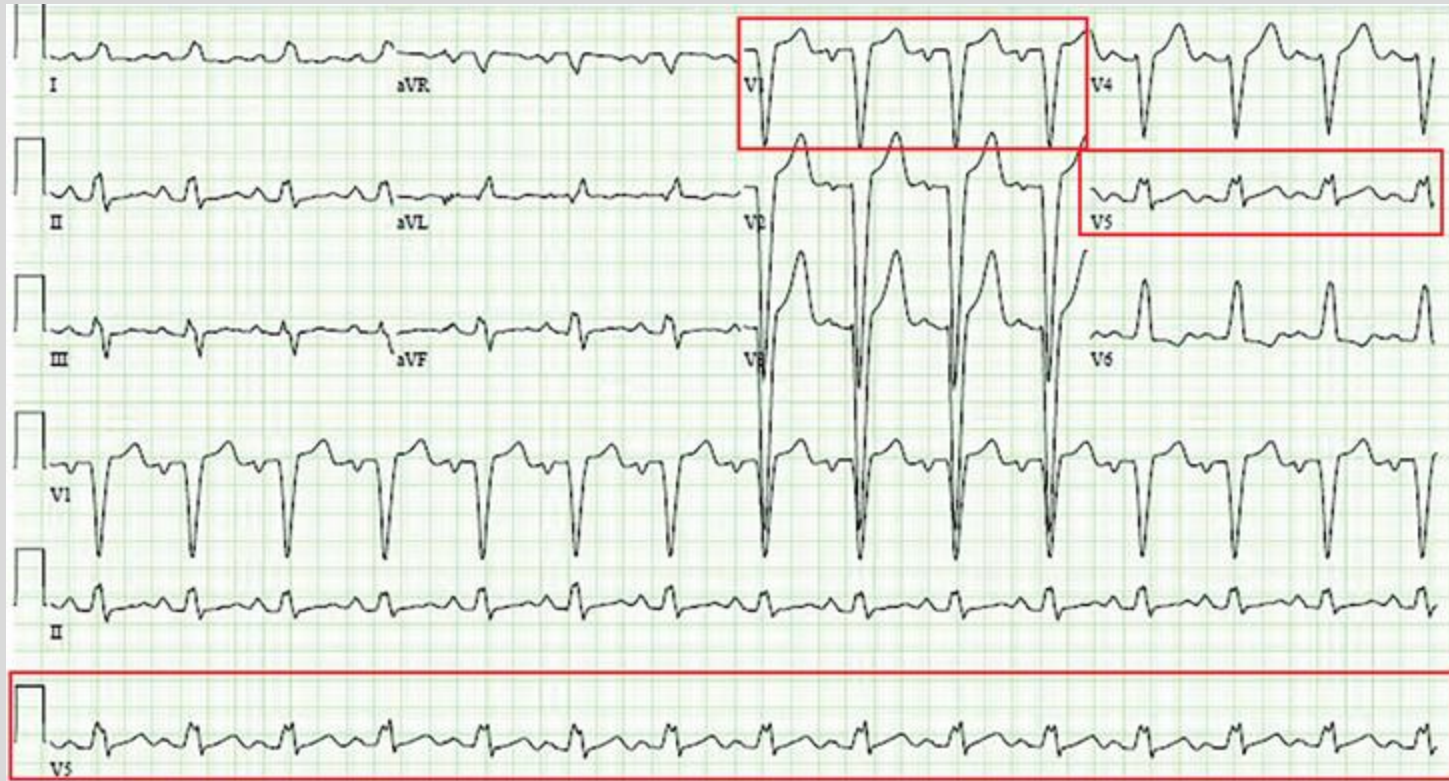


Left Bundle Branch Block (LBBB)

- Normally, the septum is activated from left to right
- In LBBB, septal depolarization is **reversed**
- Leads to **QRS duration > 120ms**
 - Produces tall R waves in the lateral leads (I, V5-6) and deep S waves in the right precordial leads (V1-3)
- Produces a broad or notched ('M'-shaped) R wave in the lateral leads (V5, V6) and ('W'-shaped) dominant deep S waves in V1

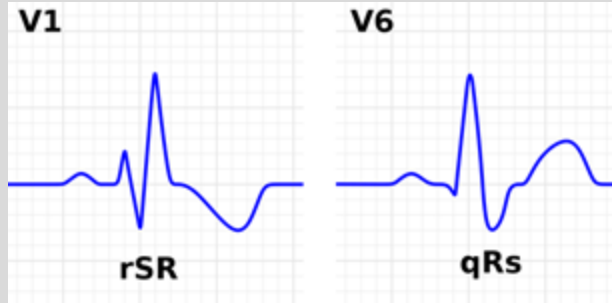


Left Bundle Branch Block (LBBB)



Right Bundle Branch Block (RBBB)

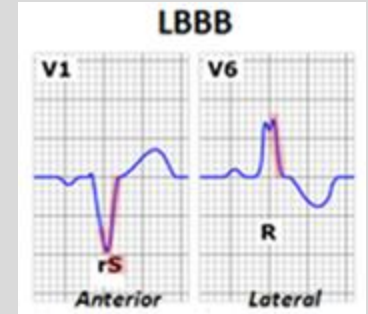
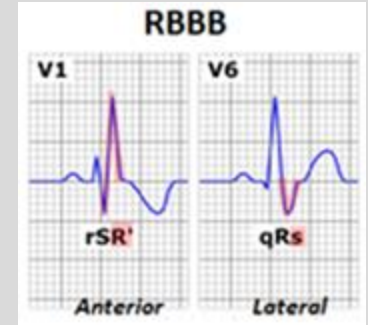
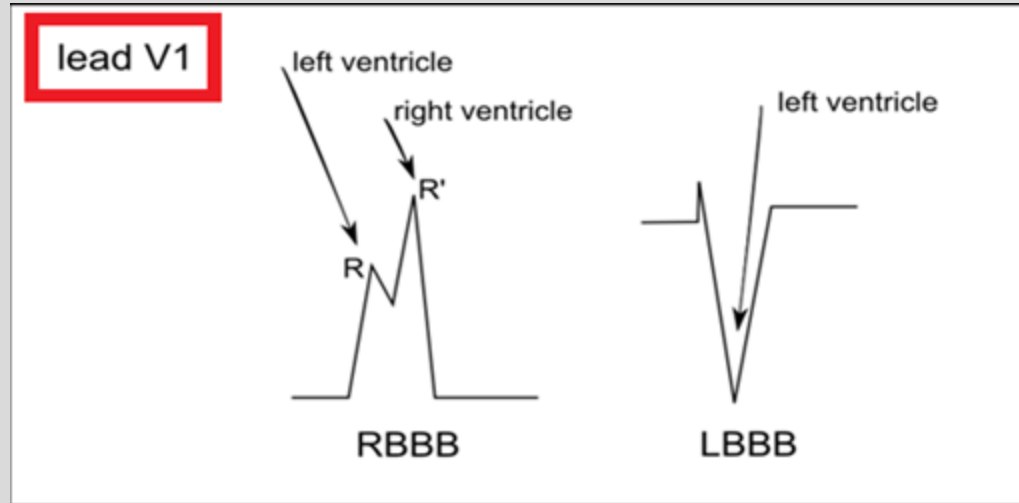
- QRS duration is greater than 120 msec
- QRS in **V1-V2 displays rSR' pattern** “bunny ears” or the letter “M”
- Lateral leads (I, V6) broad deep S-wave that is greater than R-wave duration.





Bundle Branch Block “Turn Signal Theory”

Criteria: QRS complex > 120 msec, **reference lead V1**



Summary of Objective #2

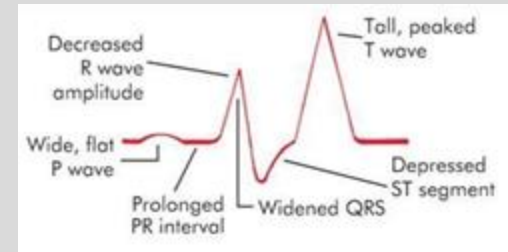
- LBBB vs. RBBB: reference lead V1 with the “Turn Signal Theory”
- LBBB: ‘W’-shaped in V1
- RBBB: ‘M’-shaped in V1

Wanna talk electrolyte disturbances?

K

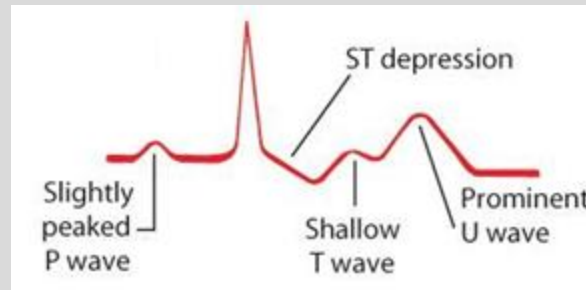
Hyperkalemia

- Potassium (K⁺): 3.5-5.0 mEq/L
- ECG changes depend on serum level and how rapid of a change
- K⁺ = 5.5 - 6.5 mEq/L
 - ****Tall, peaked, narrow based T waves****
 - QT interval shortening
- K⁺ = 6.5 - 7.5 mEq/L
 - First-degree AV block
 - **QRS widening**
- K⁺ > 7.5 mEq/L
 - Disappearance of P waves



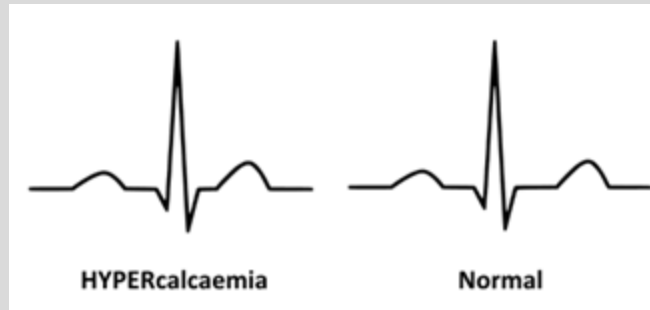
Hypokalemia (Hypokalemia)

- Severe or symptomatic hypokalemia
 - Depression of the ST segment, decrease in the amplitude of the T-wave, and an increase in the amplitude of U waves
 - U-waves are often seen in the lateral precordial leads V4 to V6
 - Prolongs the QT interval
 - Increased amplitude and width of the P-wave



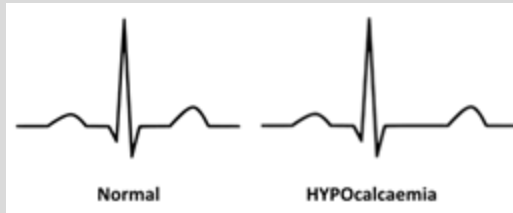
Hypercalcemia

- Calcium, serum (total): 8.5-10.2 mg/dL
- **QT interval shortening** (usually due to shortening of the ST segment)
- Note: Little if any effect on P, QRS, or T wave



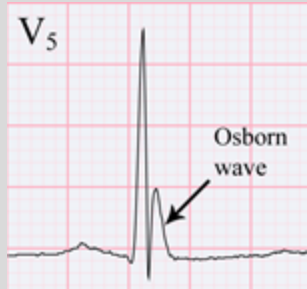
Hypocalcemia

- **Prolonged QT interval** (earliest and most common finding) due to ST segment prolongation
- No change in the duration of the T wave (seen only with hypocalcemia or hypothermia)



Hypothermia: Osborne “J” Waves

- Hypothermia results in sinus bradycardia with widening of the QRS
- Prolongation of PR and QT intervals
- **Osborne (“J”) waves**, which are late upright terminal deflections of the QRS complex (“camel hump” sign)



Summary

- Hyperkalemia: tall peaked T-waves, QT interval shortening
- Hypokalemia: ST segment depression, shallow T-waves, U-waves
- Hypercalcemia: QT interval shortening
- Hypocalcemia: Prolonged QT interval
- Hypothermia: Osborne “J” waves

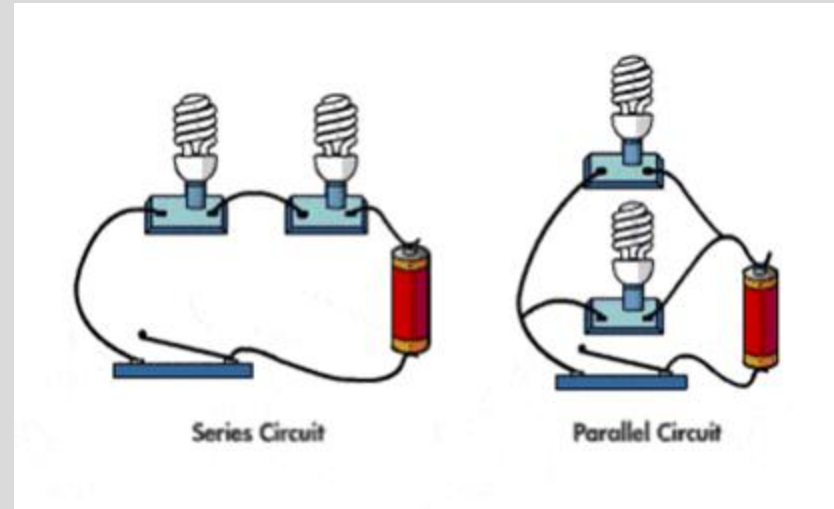
More “fun” topics:

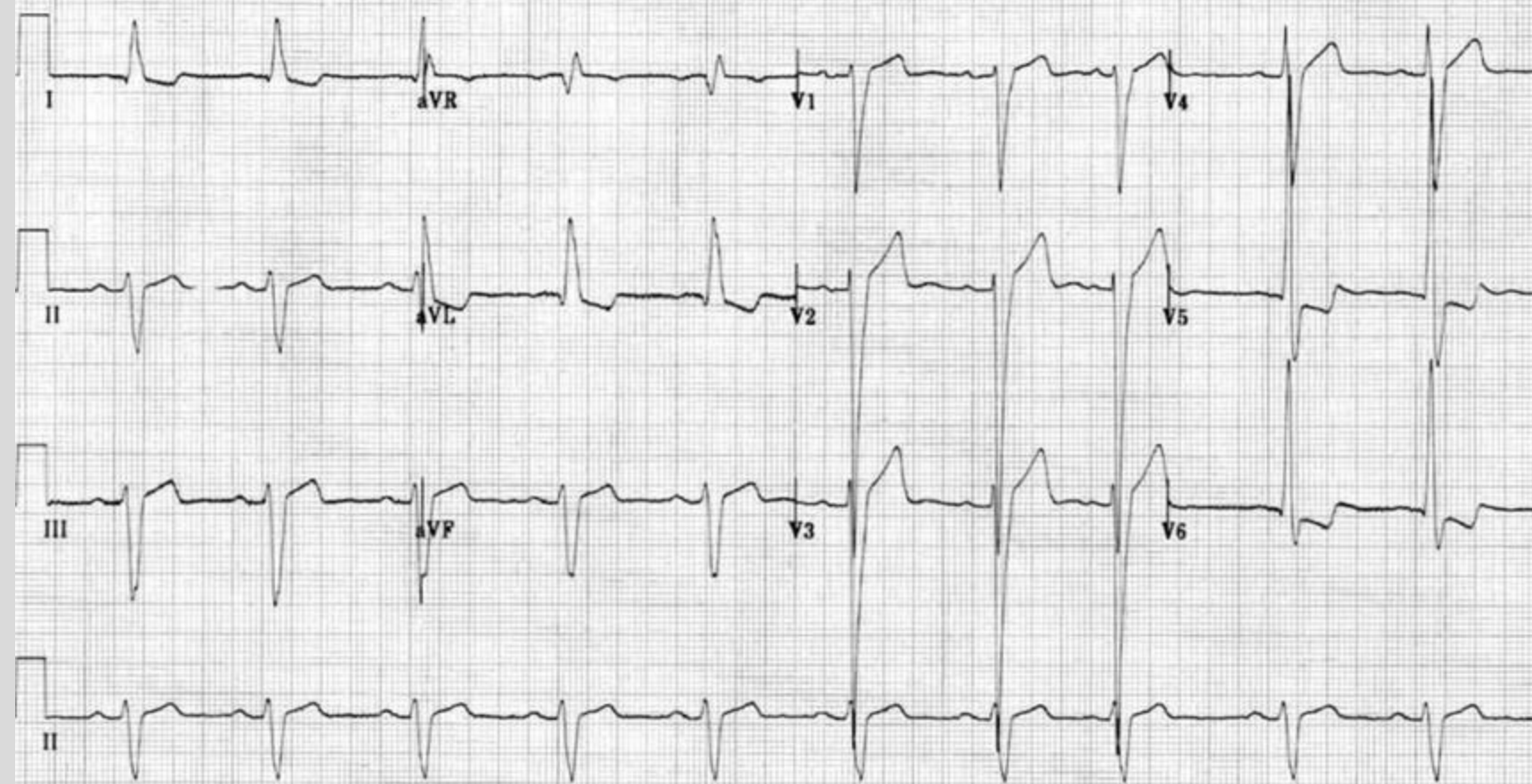
Left Ventricular Hypertrophy

Sick Sinus Syndrome

Left Ventricular Hypertrophy (LVH)

- Response to pressure overload states
- Increase in size of the myocardial fibers
 - **Parallel addition of sarcomeres**
- Causes
 - Hypertension (most common cause)
 - Aortic stenosis
 - Aortic regurgitation
 - Mitral regurgitation
 - Coarctation of the aorta
 - Hypertrophic cardiomyopathy





Sick Sinus Syndrome

- When the SA node is damaged and no longer generates normal heartbeats at a normal rate
 - Another cause of Pathologic Bradycardia
- Most common caused is gradual loss of SA node function that comes with age
- Sx: Dizziness, Pre-syncope, Syncope, SOB, Fatigue
- Tx: Pacemaker

How I would approach the lecture:

1. Differentiate and manage the different types of AV blocks.
 - a. What is an AV block?
 - b. How many AV blocks are there? How do we separate them based on tx?
 - c. What are the ECG findings of the different AV blocks?

How I would approach the lecture:

2. Understand bundle anatomy and identify between left vs right bundle branch block.
 - a. Know what the bundle branches are.
 - b. Understand that the QRS will be prolonged.
 - c. Identify the ECG findings of the L and R BBBs.

How I would approach the lecture:

3. Recognize ECG changes due to electrolyte imbalances.

- a. Hypokalemia, Hyperkalemia
- b. Hypocalcemia, Hypercalcemia
- c. Osborne J Waves

Final Slide:

Questions:

Ermin Tale: ermin.tale01@gmail.com

Justin Edward: jedward@nyit.edu

Grace Robinson: grobin03@nyit.edu

More exposure → better EKG skills

-ECG Lectures **Mon 12-1 PM EST** with Dr. Cohen via Zoom -Will be sent out to the class each week by Dr. Cohen's Research Team

Feedback on the lecture:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>